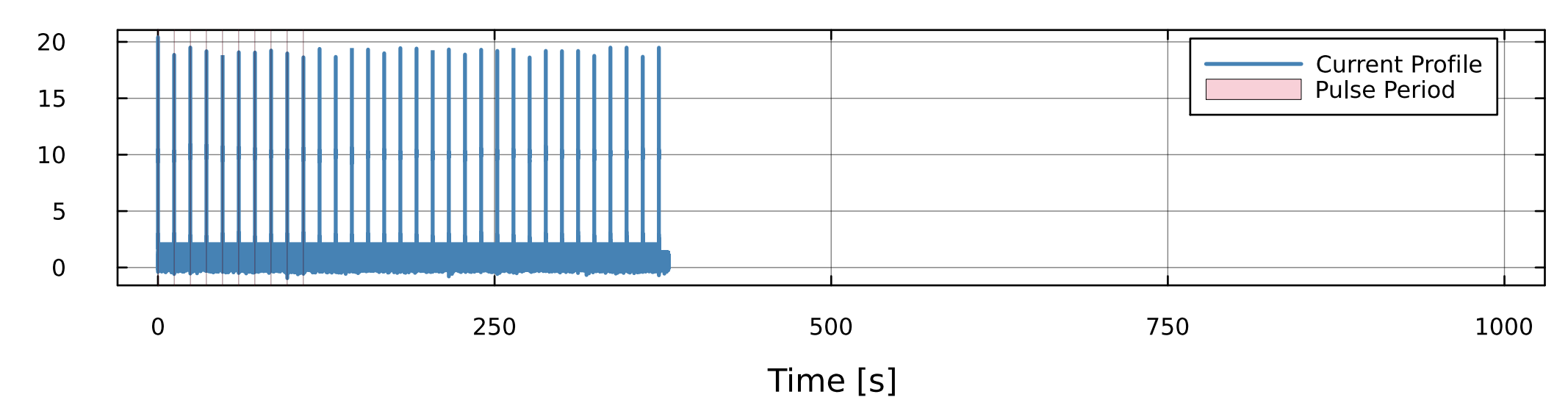


Frequency Pulse Current Analysis: From External Excitation to Internal Mechanistic Narrative of Battery Response Under Tesla-Style Fast Charging

(a) High-Frequency Pulse Current Excitation



(b) Cell Voltage Response

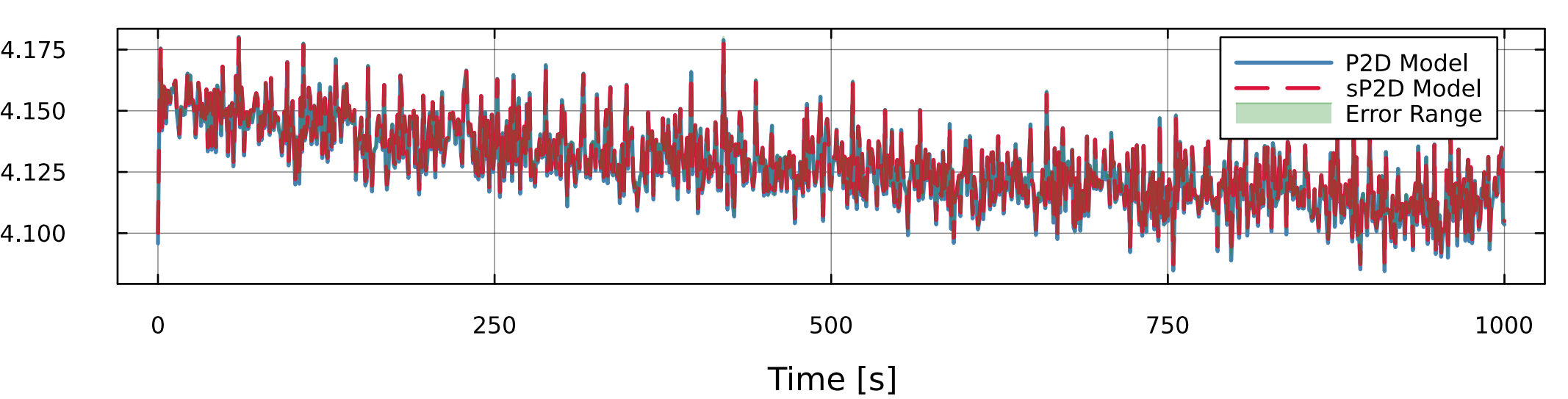
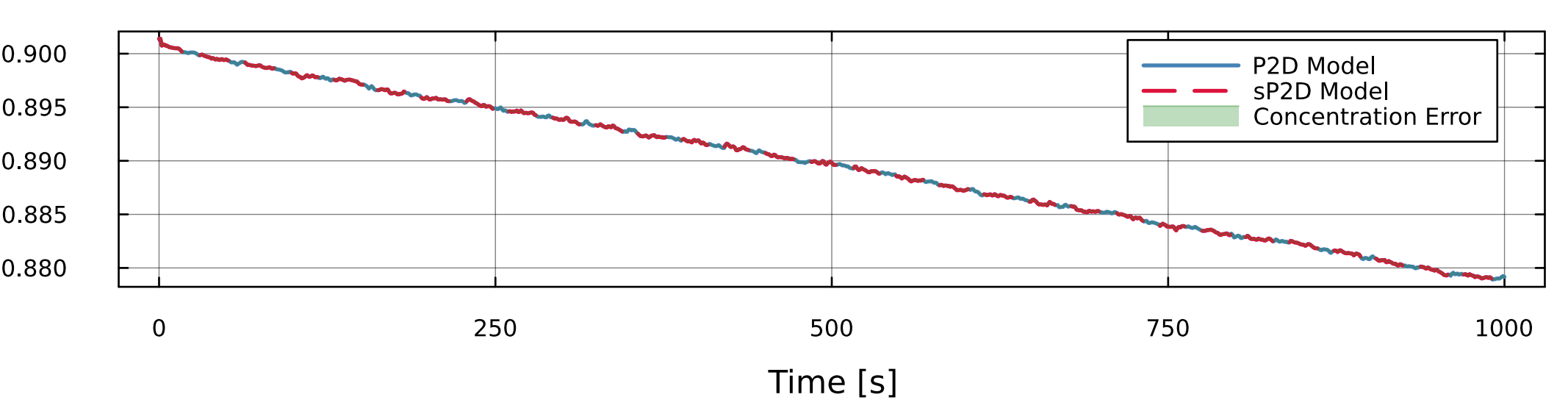


Figure 2: Internal State Evolution Mechanism
Concentration gradients and overpotential dynamics drive electrochemical

(a) Lithium Concentration Evolution



(b) Electrode Overpotential Dynamics

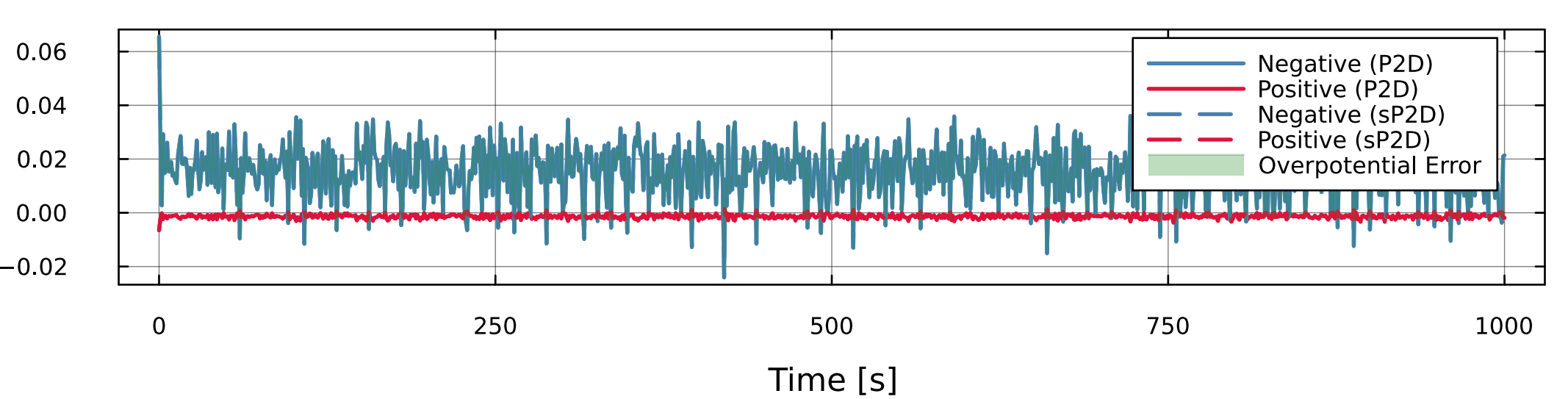
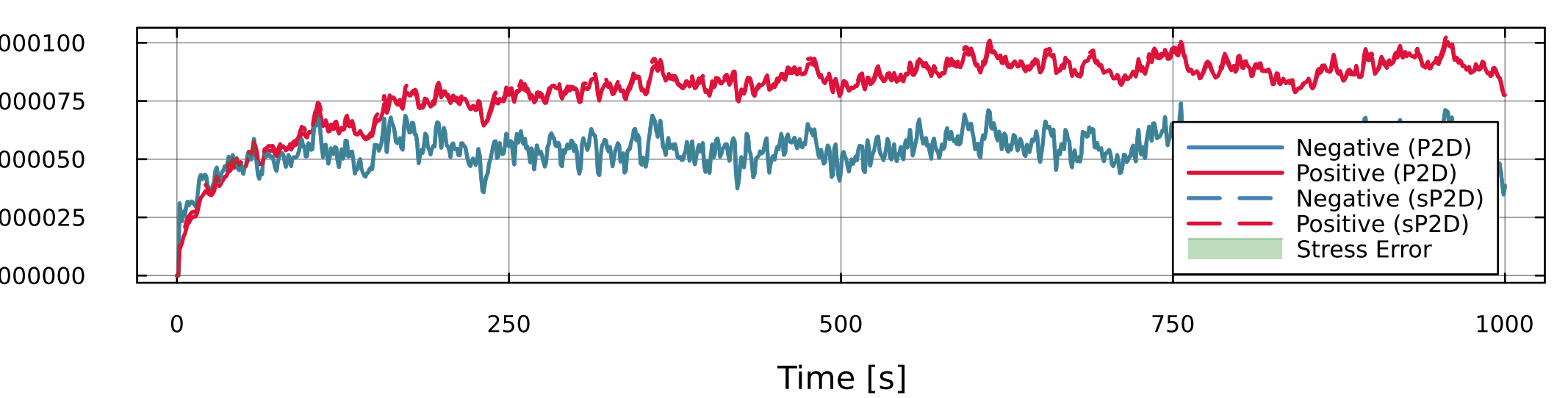


Figure 3: Stress Distribution and Error Analysis
al stress evolution and model prediction accuracy under high-freque

(a) Mechanical Stress Evolution



(b) Model Prediction Errors

